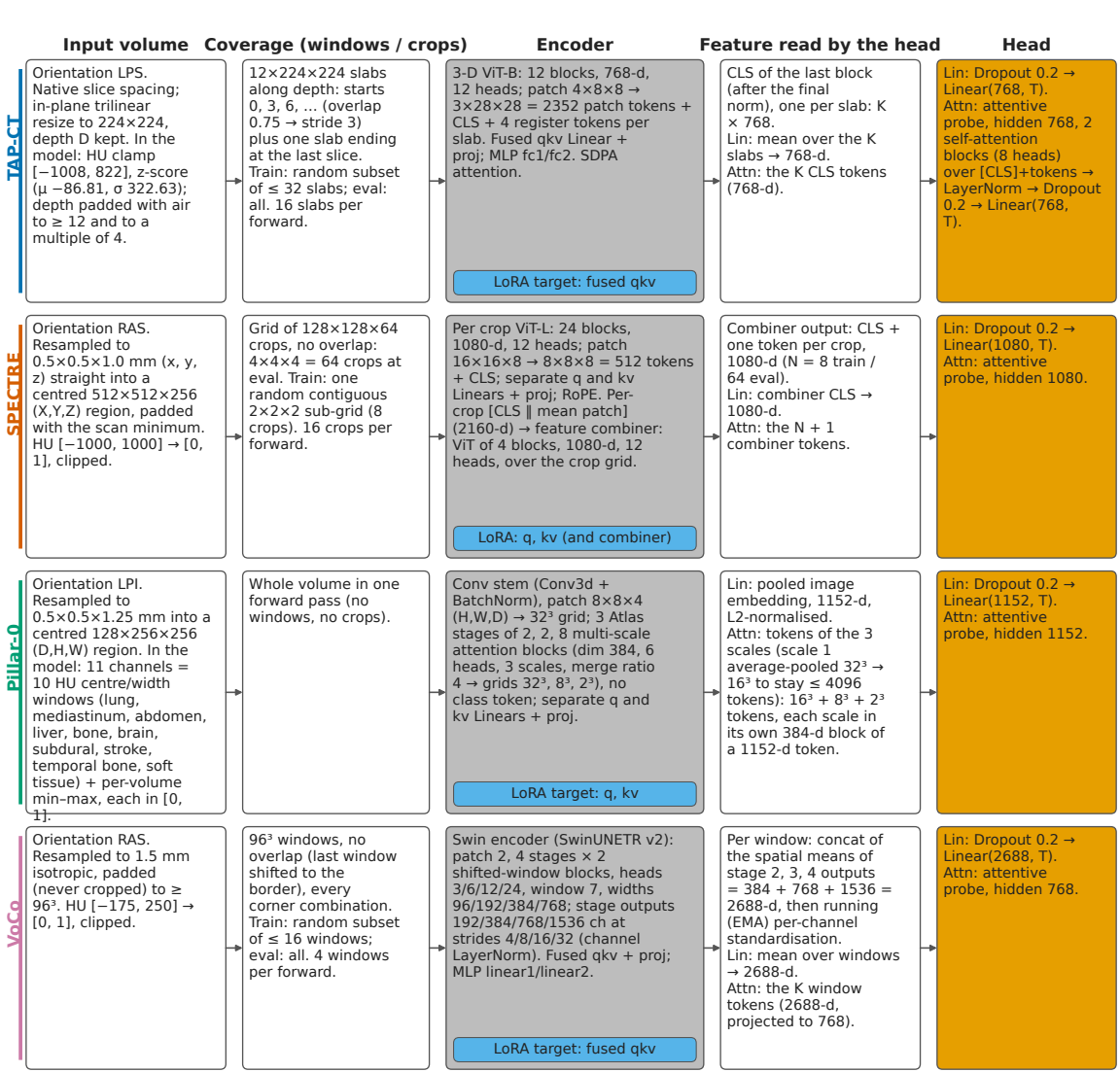
Input volume
Coverage (windows / crops)
Encoder
Feature read by the head
Head
TAP-CT
Orientation LPS. Native slice spacing; in-plane trilinear resize to 224×224, depth D kept. In the model: HU clamp [−1008, 822], z-score (μ −86.81, σ 322.63); depth padded with air to ≥ 12 and to a multiple of 4.
12×224×224 slabs along depth: starts 0, 3, 6, … (overlap 0.75 → stride 3) plus one slab ending at the last slice. Train: random subset of ≤ 32 slabs; eval: all. 16 slabs per forward.
3-D ViT-B: 12 blocks, 768-d, 12 heads; patch 4×8×8 → 3×28×28 = 2352 patch tokens + CLS + 4 register tokens per slab. Fused qkv Linear + proj; MLP fc1/fc2. SDPA attention.
LoRA target: fused qkv
CLS of the last block (after the final norm), one per slab: K × 768. Lin: mean over the K slabs → 768-d. Attn: the K CLS tokens (768-d).
Lin: Dropout 0.2 → Linear(768, T). Attn: attentive probe, hidden 768, 2 self-attention blocks (8 heads) over [CLS]+tokens → LayerNorm → Dropout 0.2 → Linear(768, T).
SPECTRE
Orientation RAS. Resampled to 0.5×0.5×1.0 mm (x, y, z) straight into a centred 512×512×256 (X,Y,Z) region, padded with the scan minimum. HU [−1000, 1000] → [0, 1], clipped.
Grid of 128×128×64 crops, no overlap: 4×4×4 = 64 crops at eval. Train: one random contiguous 2×2×2 sub-grid (8 crops). 16 crops per forward.
Per crop ViT-L: 24 blocks, 1080-d, 12 heads; patch 16×16×8 → 8×8×8 = 512 tokens + CLS; separate q and kv Linears + proj; RoPE. Per-crop [CLS ‖ mean patch] (2160-d) → feature combiner: ViT of 4 blocks, 1080-d, 12 heads, over the crop grid.
LoRA: q, kv (and combiner)
Combiner output: CLS + one token per crop, 1080-d (N = 8 train / 64 eval). Lin: combiner CLS → 1080-d. Attn: the N + 1 combiner tokens.
Lin: Dropout 0.2 → Linear(1080, T). Attn: attentive probe, hidden 1080.
Pillar-0
Orientation LPI. Resampled to 0.5×0.5×1.25 mm into a centred 128×256×256 (D,H,W) region. In the model: 11 channels = 10 HU centre/width windows (lung, mediastinum, abdomen, liver, bone, brain, subdural, stroke, temporal bone, soft tissue) + per-volume min–max, each in [0, 1].
Whole volume in one forward pass (no windows, no crops).
Conv stem (Conv3d + BatchNorm), patch 8×8×4 (H,W,D) → 32^3 grid; 3 Atlas stages of 2, 2, 8 multi-scale attention blocks (dim 384, 6 heads, 3 scales, merge ratio 4 → grids 32^3, 8^3, 2^3), no class token; separate q and kv Linears + proj.
LoRA target: q, kv
Lin: pooled image embedding, 1152-d, L2-normalised. Attn: tokens of the 3 scales (scale 1 average-pooled 32^3 → 16^3 to stay ≤ 4096 tokens): $16^3 + 8^3 + 2^3$ tokens, each scale in its own 384-d block of a 1152-d token.
Lin: Dropout 0.2 → Linear(1152, T). Attn: attentive probe, hidden 1152.
VoCo
Orientation RAS. Resampled to 1.5 mm isotropic, padded (never cropped) to ≥ 96^3. HU [−175, 250] → [0, 1], clipped.
96^3 windows, no overlap (last window shifted to the border), every corner combination. Train: random subset of ≤ 16 windows; eval: all. 4 windows per forward.
Swin encoder (SwinUNETR v2): patch 2, 4 stages × 2 shifted-window blocks, heads 3/6/12/24, window 7, widths 96/192/384/768; stage outputs 192/384/768/1536 ch at strides 4/8/16/32 (channel LayerNorm). Fused qkv + proj; MLP linear1/linear2.
LoRA target: fused qkv
Per window: concat of the spatial means of stage 2, 3, 4 outputs = 384 + 768 + 1536 = 2688-d, then running (EMA) per-channel standardisation. Lin: mean over windows → 2688-d. Attn: the K window tokens (2688-d, projected to 768).
Lin: Dropout 0.2 → Linear(2688, T). Attn: attentive probe, hidden 768.
T = number of outputs (1 per-task, K joint). LoRA+ABMIL: gated-attention MIL pooling (hidden 128) over the token set, then the probe on the pooled token. Encoder: frozen under Lin/Attn, adapted or trained under the other methods.
frozen (pretrained weights)
low-rank adapter, trained
trained in full